Supplementary material: Modelling Pro-Rata Constraints in Power Systems Using Mixed Integer Linear Programming

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I. IEEE 24 Bus Testcase Data

TABLE I: Technical Data of Generating Units

ID	Bus	Pro-Rata Group	$\underline{P}_g^G$ [MW]	$\hat{P}_g^G$ [MW]	$\overline{P}_g^G$ [MW]
G1	1		30.4	152	152
G2	2		30.4	152	152
G3	7		75	350	350
G4	13		206.85	591	591
G5	15		12	60	60
G6	15		54.25	155	155
G7	16		54.25	155	155
G8	18		100	400	400
G9	21		100	400	400
G10	22		300	300	300
G11	23		108.5	310	310
G12	23		140	350	350
W1.1	1	South, {1,3}	0	0	300
W1.2	1	South, {1,3}	0	0	300
W2	2	South, {2,5,7}, {2,3,5,7}	0	0	300
W3	3	South, {1,3}, {2,3,5,7}	0	0	300
W5	5	South, {2,5,7}, {2,3,5,7}	0	0	300
W7	7	South, {2,5,7}, {2,3,5,7}	0	0	300

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TABLE II: Cost Data of Generating Units

ID	c_g^{G1} [mu]	c_g^{G0} [mu/MWh]	$c_g^{\mathrm{G}\downarrow F}$ [mu/MWh]	$c_g^{\mathrm{G}\uparrow F}$ [mu/MWh]
G1	13.32	1430.4	11	15
G2	13.32	1430.4	11	15
G3	20.7	1725	16	24
G4	20.93	3056.7	17	25
G5	26.11	437	23	28
G6	10.52	312	7	16
G7	10.52	312	7	16
G8	6.02	0	50	6.02
G9	5.47	0	50	5.47
G10	0	0	0	0
G11	0	624	8	14
G12	0	2298	8	16
W1.1	0	0	50	0
W1.2	0	0	50	0
W2	0	0	50	0
W3	0	0	50	0
W5	0	0	50	0
W7	0	0	50	0

TABLE III: Modified Load Data

ID	Bus	Demand [% of Total]	Demand [MW]
D1	1	1.9	33.74
D2	2	1.7	30.19
D3	3	3.2	56.83
D4	4	1.3	23.09
D5	5	1.3	23.09
D6	6	2.4	42.62
D7	7	2.2	39.07
D8	8	3.0	53.28
D9	9	3.1	55.05
D10	10	3.4	60.38
D11	13	13.4	237.96
D12	14	9.8	174.03
D13	15	16.0	284.13
D14	16	5.0	88.79
D15	18	16.8	298.34
D16	19	9.0	159.83
D17	20	6.5	115.43

TABLE IV: Technical Data of Prosumers

ID	Bus	$\overline{P}_s^{S+}, \hat{P}_s^{S+}*$ [MW]	$\overline{P}_s^{S-}, \hat{P}_s^{S-}*$ [MW]	$\underline{P}_s^{S+}, \underline{P}_s^{S-}$ [MW]	P_s^S [MW]
P1	1	1000	-1000	0	0.1
P2	2	1000	-1000	0	0.1
P3	3	1000	-1000	0	0.1
P5	5	1000	-1000	0	0.1
P7	7	1000	-1000	0	0.1
P11	11	1000	-1000	0	-200
P12	12	1000	-1000	0	-200
P24	24	1000	-1000	0	-200

**Note:* $\overline{P}_s^{S-}, \hat{P}_s^{S-}, \underline{P}_s^{S+}, \underline{P}_s^{S-}$ are varied within the prosumer trials. These data apply for the snapshot trial D04.